

Mobile Application Development

Lecture 03

Today's Agenda

- Running the Android App Using Emulator
- Java Virtual machine (JVM) vs (Dalvic VM)DVM vs (Android RunTime)ART

Open the Android Virtual Device Manager

The AVD Manager allows you to set up new AVDs, and view and edit ones you've already created. Open it by selecting Android on the Tools menu and choosing AVD Manager.

If you have no AVDs set up already, you'll be presented with a screen prompting you to create one. Click on the "Create Virtual Device" button.

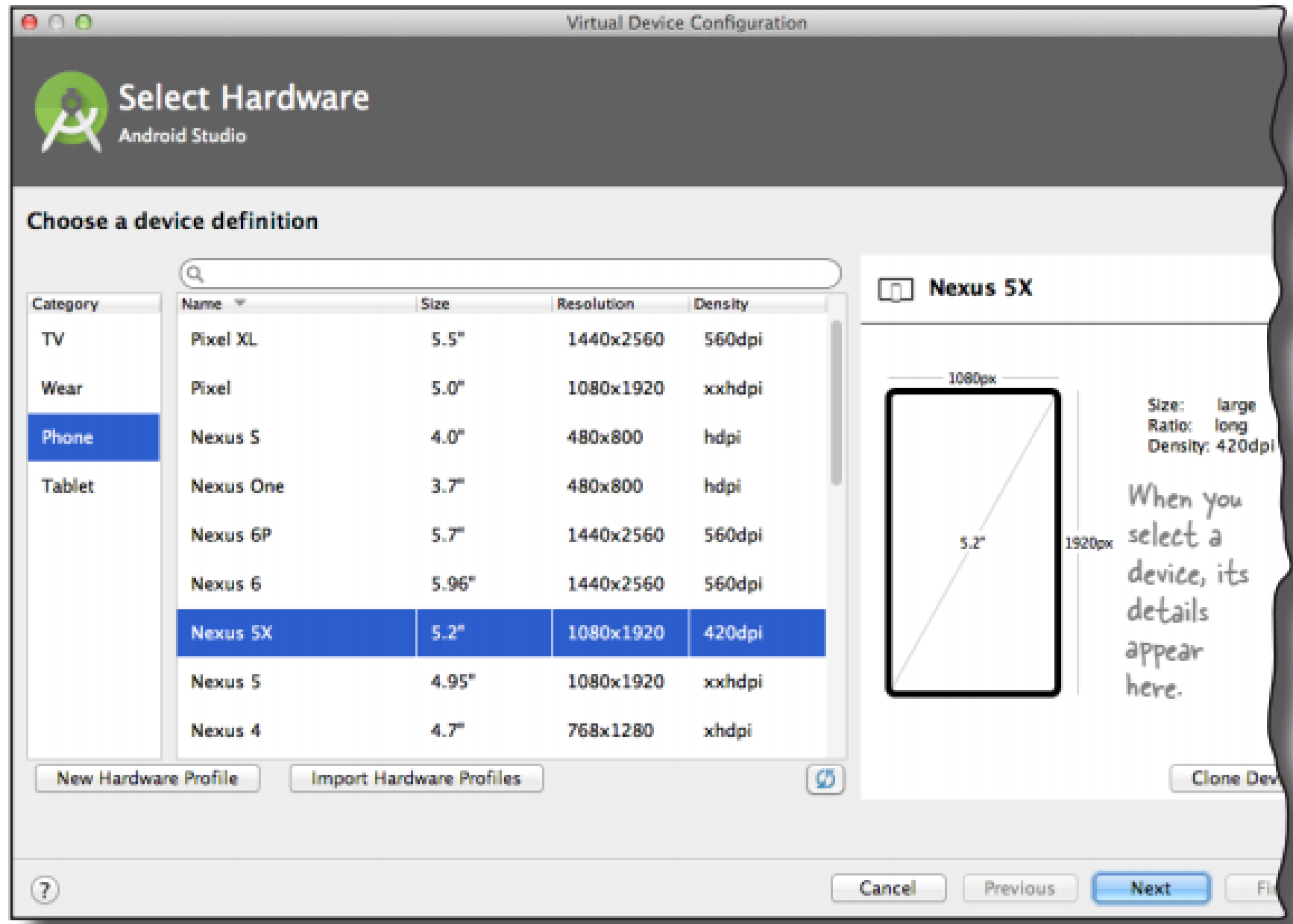
*Click on this button
to create an AVD.*



Select the hardware

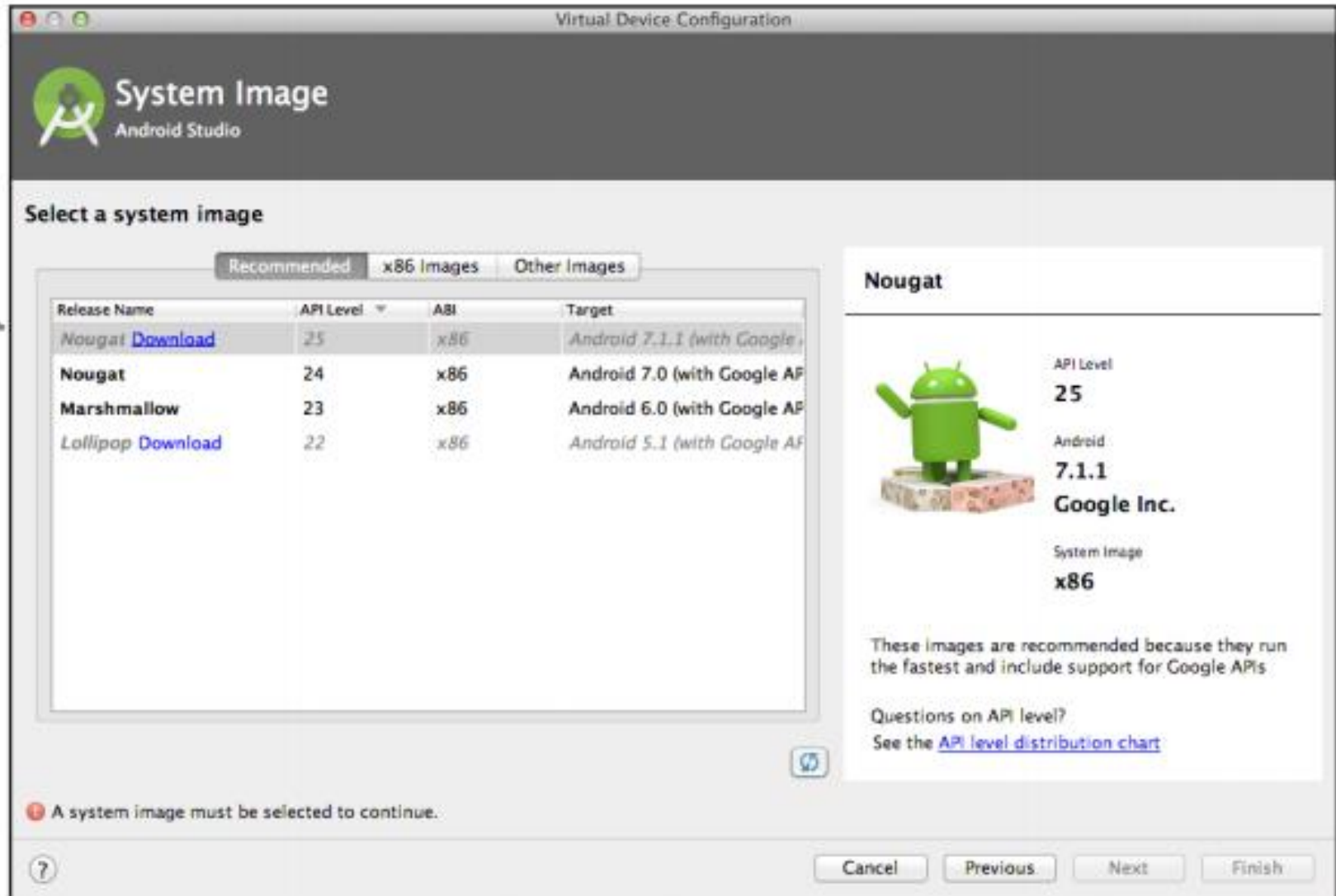
On the next screen, you'll be prompted to choose a device definition. This is the type of device your AVD will emulate. You can choose a variety of phone, tablet, wear, or TV devices.

We're going to see what our app looks like running on a Nexus 5X phone. Choose Phone from the Category menu and Nexus 5X from the list. Then click the Next button.

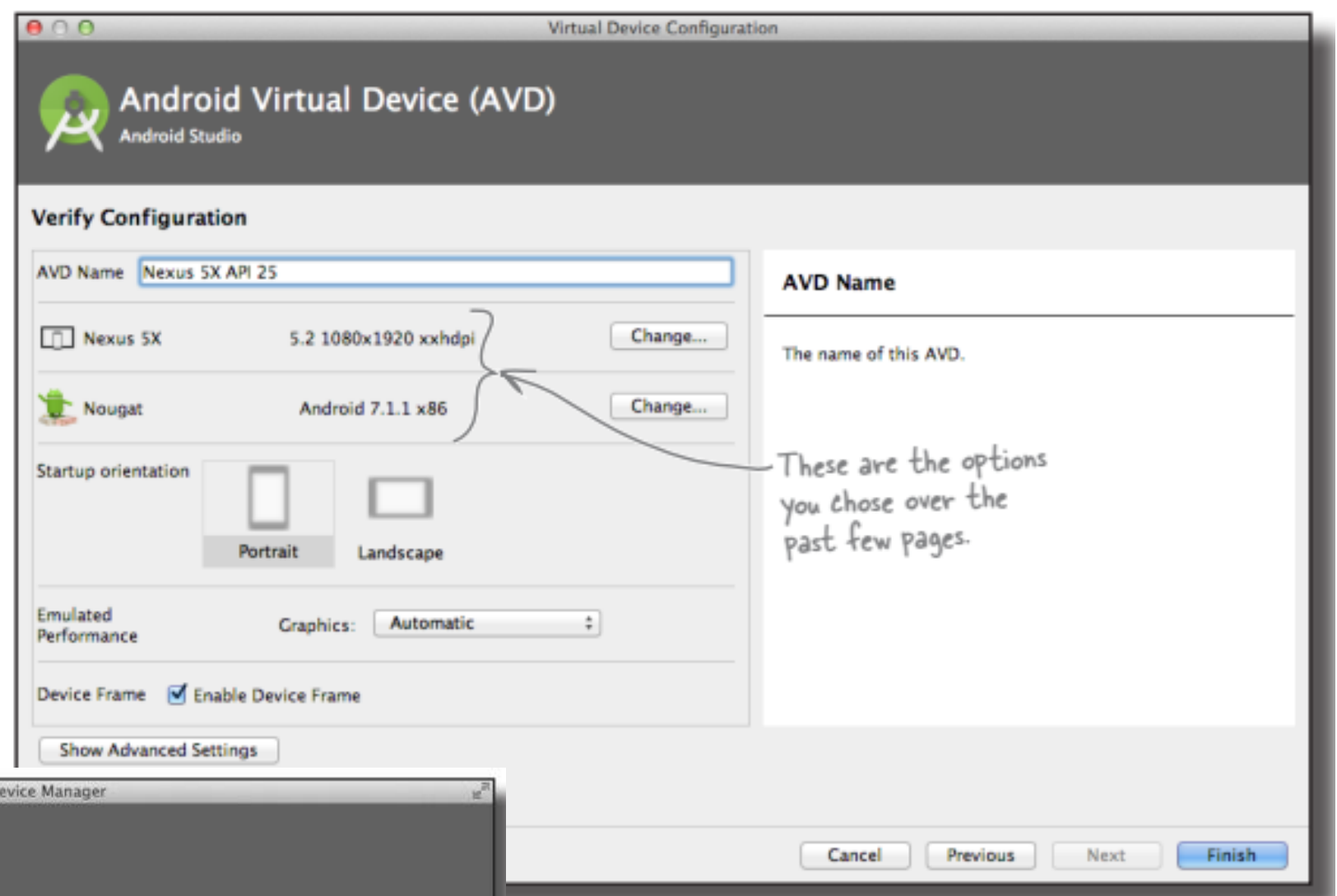


Select a system image

If you don't have this system image installed, you'll be given the option to download it.

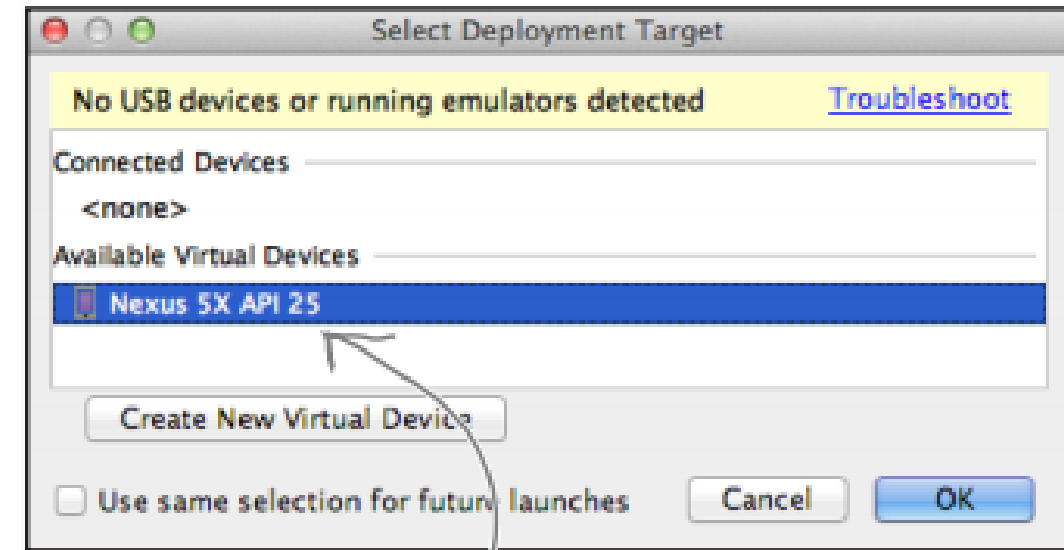
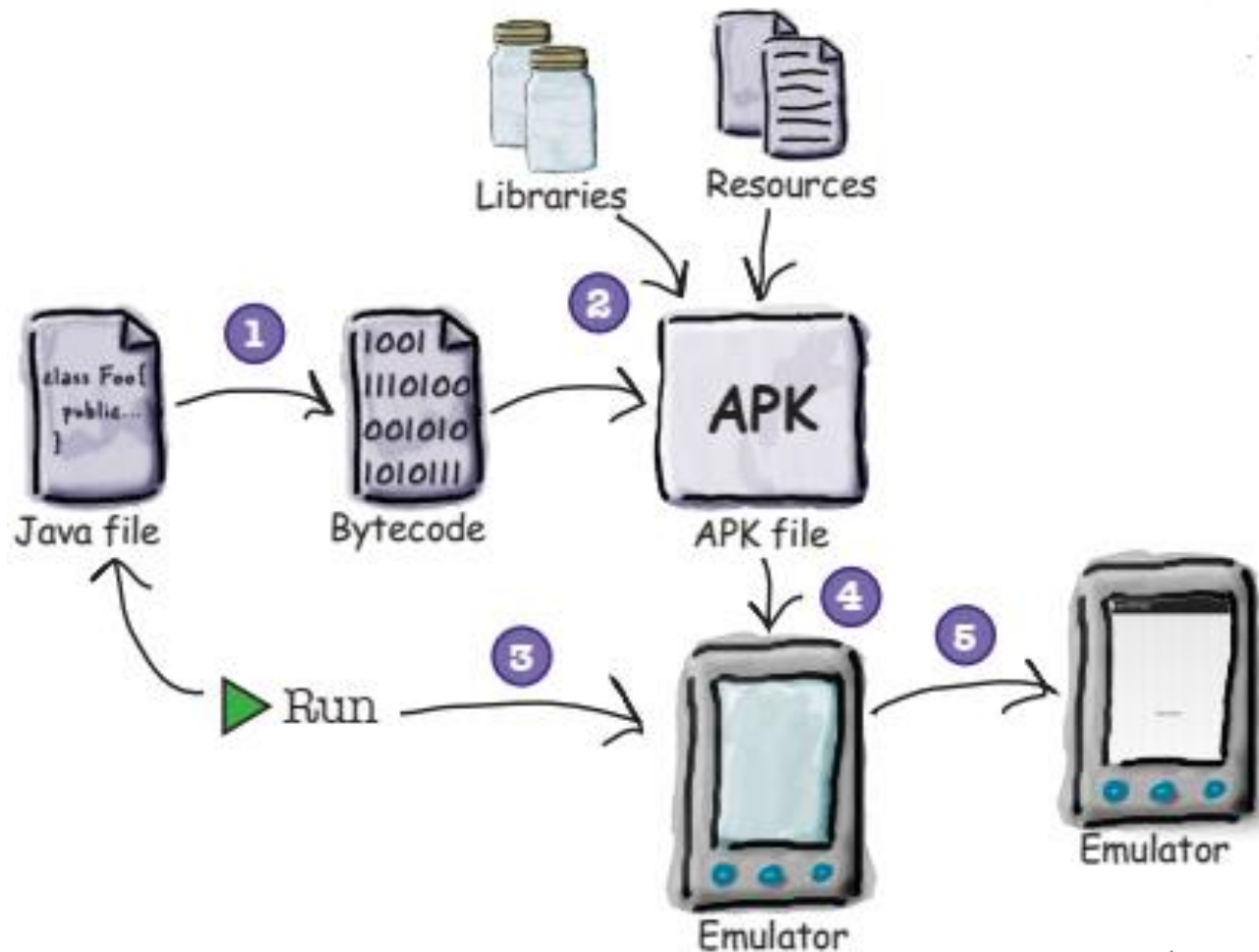


Verify the AVD configuration



Run the app in the emulator

- Compile, package, deploy, and run



This is the AVD we just created.

- 1** The Java source files get compiled to bytecode.
- 2** An Android application package, or APK file, gets created.
The APK file includes the compiled Java files, along with any libraries and resources needed by your app.
- 3** Assuming there's not one already running, the emulator gets launched and then runs the AVD.
- 4** Once the emulator has been launched and the AVD is active, the APK file is uploaded to the AVD and installed.
- 5** The AVD starts the main activity associated with the app.
Your app gets displayed on the AVD screen, and it's all ready for you to test out.

You can watch progress in the console

03/13 10:45:41: Launching app

```
$ adb install-multiple -r /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/intermediates/split-apk/debug/slices/slice_1.apk /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/intermediates/split-apk/debug/slices/slice_2.apk /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/intermediates/split-apk/debug/slices/slice_0.apk /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/intermediates/split-apk/debug/slices/slice_3.apk /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/intermediates/split-apk/debug/slices/slice_6.apk /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/intermediates/split-apk/debug/slices/slice_4.apk /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/intermediates/split-apk/debug/slices/slice_5.apk /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/intermediates/split-apk/debug/slices/slice_7.apk /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/intermediates/split-apk/debug/slices/slice_8.apk /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/intermediates/split-apk/debug/slices/slice_9.apk /Users/dawng/AndroidStudioProjects/MyFirstApp/app/build/outputs/apk/app-debug.apk
```

Split APKs installed

```
$ adb shell am startservice com.hfad.myfirstapp/com.android.tools.fd.runtime.InstantRunService
```

```
$ adb shell am start -n "com.hfad.myfirstapp/com.hfad.myfirstapp.MainActivity" -a android.intent.action.MAIN -c android.intent.category.LAUNCHER
```

```
Connected to process 2685 on device Nexus_5X_API_25 [emulator-5554]
```

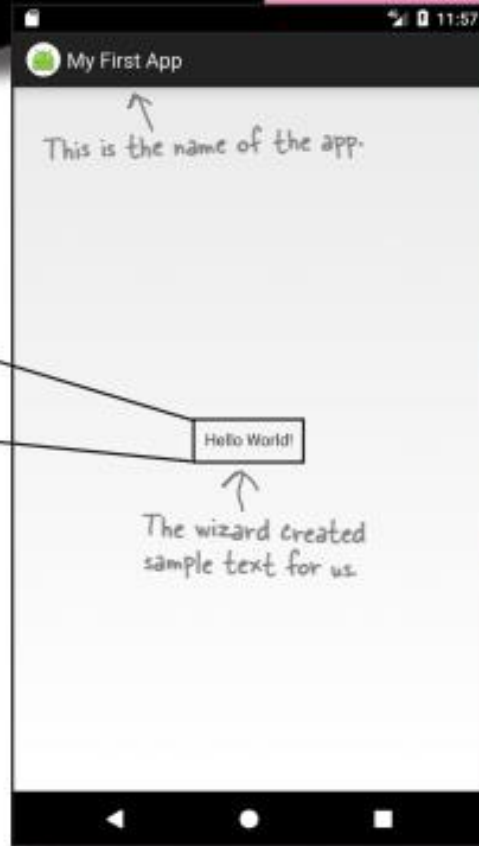
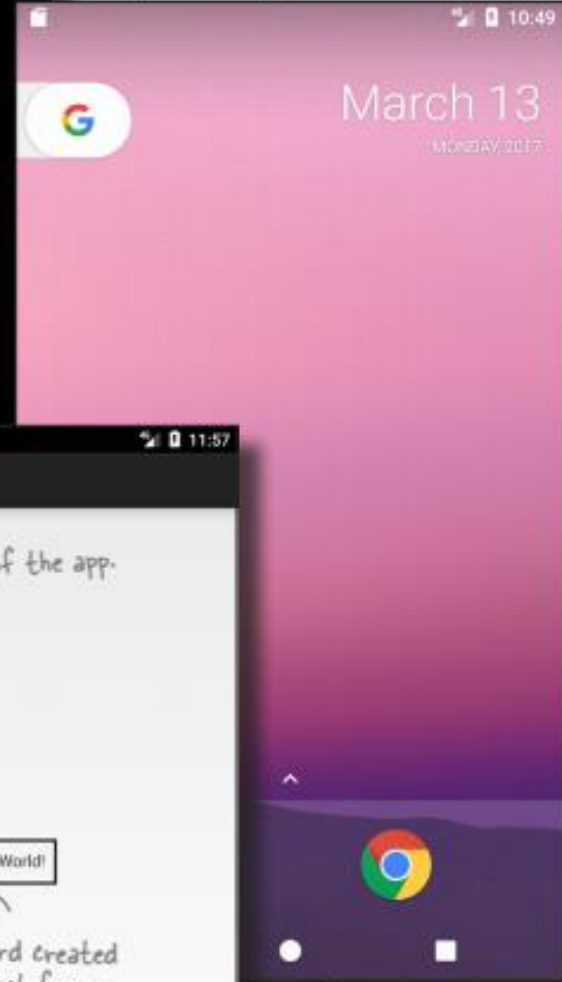


Android Studio has finished launching the AVD we just set up.

The emulator launches our app by starting the main activity for it. This is the activity the wizard created for us.

The emulator launches...

...and here's the AVD home screen. It looks and behaves just like a real Nexus 5X device.



Hello World!

Android Studio created the sample text "Hello World!" without us telling it to.

Hello World!

The wizard created sample text for us.

Here's the app running on the AVD.